Transportation Sustainability and Vehicle CO_2 Emissions Prediction

A Predictive Analytics Study Using Canadian Fuel Consumption Data

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Updated Milestone 1

1. Professional Context and Problem Definition

1.1 Selected Job-Simulation Track

This project is related to transportation sustainability and vehicle emissions analysis. The project is designed from the perspective of a transportation sustainability analyst who uses vehicle emissions data to support environmental planning and sustainability-related transportation decisions.

The project examines how various vehicle characteristics affect fuel consumption and CO2 emissions using predictive analytics.

1.2 Problem or Decision Context

Transportation sustainability analysts need data-driven insights to identify which vehicle characteristics are associated with higher fuel consumption and CO2 emissions. This information can support sustainability planning and help evaluate the environmental impact of different vehicle categories. Different vehicles produce different levels of emissions depending on factors such as engine size, cylinders, fuel type, and transmission type. This project focuses on analyzing vehicle data to identify the factors most related to fuel consumption and emissions performance.

1.3 Key Beneficiary or Target Audience

The primary beneficiary of this project is transportation sustainability analysts. The findings may also support environmental researchers and policymakers interested in vehicle emissions and fuel efficiency trends.

1.4 Real-World Significance

Transportation worldwide is a major source of greenhouse gas emissions, which can harm the environment. In this project, I will study real-world vehicle data to better understand how different vehicle factors impact the environment. Using real-world vehicle data, this project can help identify patterns related to fuel consumption and emissions and support better decisions for environmental sustainability.

2. Research Questions and Impact Hypothesis

2.1 Research Questions

- **a.** Which categories of vehicles sold in Canada between 2015 and 2024 are associated with higher fuel consumption and CO_2 emissions, and how can this information support sustainable transportation planning?
- **b.** What vehicle characteristics contribute the most to higher emissions in Canadian light-duty vehicles, and how can these insights support transportation analysts in identifying lower-emission vehicle options?
- **c.** How does fuel consumption affect CO_2 emissions in Canadian light-duty vehicles, and what does this reveal about the environmental performance of different vehicles?

Research Questions and Related Dataset Variables:

Questions	Approach	Tools
A. Which categories of vehicles sold in Canada between 2015 and 2024 are associated with higher fuel consumption and CO_2 emissions, and how can this information support sustainable transportation planning?	Compare fuel consumption and CO_2 emissions across different vehicle categories and identify high-emission vehicle groups using exploratory analysis and visualizations.	Python, Google Colab, Pandas, Matplotlib, Seaborn, Histograms, Boxplots, Grouped Analysis, Summary Statistics
B. What vehicle characteristics contribute the most to higher emissions in Canadian light-duty vehicles, and how can these insights support transportation analysts in identifying lower-emission vehicle options?	Analyze relationships between vehicle characteristics such as engine size, cylinders, fuel type, and transmission type with CO_2 emissions using correlation analysis and predictive modeling.	Correlation Matrix, Scatterplots, Multiple Linear Regression, Random Forest Regression, Feature Importance Analysis, Scikit-learn
C. How does fuel consumption affect CO_2 emissions in Canadian light-duty vehicles, and what does this reveal about the environmental performance of different vehicles?	Examine the relationship between fuel consumption measures and CO_2 emissions using descriptive analysis, visualizations, correlation analysis, and predictive modelling to understand how fuel use influences environmental performance.	Exploratory Data Analysis (EDA), Correlation Analysis, Scatterplots, Trend Analysis, Regression Models, Python Libraries

2.2 Why the questions matter

These questions are important because they help identify vehicle categories associated with higher emissions, determine which vehicle features contribute most to environmental impact, and examine how fuel consumption influences CO_2 emissions. Understanding of these relationships can deliver valuable insights into vehicle environmental performance and support sustainability-focused transportation analysis.

2.3 Impact Hypothesis

It is expected that vehicle features, such as engine size, cylinder count, fuel type, and transmission type, will be associated with fuel consumption and CO_2 emissions. Vehicles with larger engines and higher fuel consumption are expected to produce higher emissions compared with more fuel-efficient vehicles.

If these relationships can be identified successfully, the predictive models developed in this project should be able to estimate CO_2 emissions with reasonable accuracy. The findings may also help improve understanding of the factors that contribute most to vehicle emissions and fuel efficiency.

3. Dataset, Data Biography, and Ethics Review

3.1 Dataset Description

This project uses the Government of Canada Fuel Consumption Ratings dataset for model years 2015–2024. The dataset contains approximately 10,060 vehicle records and 15 variables describing vehicle characteristics, fuel consumption measures, environmental ratings, and CO_2 emissions. Variables include vehicle class, engine size, cylinders, transmission type, fuel type, city fuel consumption, highway fuel consumption, combined fuel consumption, CO_2 Rating, Smog Rating, and CO_2 Emissions (g/km). The CO_2 emission dataset is provided in CSV format and is suitable for exploratory data analysis and predictive modeling

3.2 Source and Provenance Trail

The dataset was obtained directly from the Government of Canada Open Data Portal and is published by Natural Resources Canada (NRCan). The dataset provides model-specific fuel consumption ratings and estimated CO_2 emissions for new light-duty vehicles sold in Canada. The selected dataset covers vehicle model years from 2015 to 2024 and is available in CSV format. The dataset is distributed under the Open Government License – Canada, which permits educational and research use. The information provides a solid basis for examining trends in vehicle fuel consumption and emissions because it is an official government source.

3.3 Relevant Characteristics and Limitations

The dataset includes both numerical and categorical variables describing vehicle specifications, fuel consumption measurements, environmental ratings, and CO_2 emissions. During the initial review of the data, approximately 10,060 vehicle records and 15 variables were identified for analysis.

Although the dataset provides detailed information about vehicle characteristics, it does not include many real-world factors that may influence fuel consumption and emissions. Variables such as driving behaviour, traffic conditions, weather, road conditions, and maintenance history are not available. As a result, any patterns or predictions developed in this project will be based only on the information contained in the dataset and should be interpreted within that context.

3.4 Privacy and Ethics Audit

The dataset used in this project contains only vehicle-related information and does not include any personally identifiable information (PII), confidential information, or sensitive individual data. The dataset consists of vehicle specifications, fuel consumption measures, environmental ratings, and CO_2 emissions data for vehicles sold in Canada. Therefore, the privacy risk associated with this project is considered low.

The dataset was obtained from the Government of Canada Open Data Portal and is available under the Open Government Licence – Canada. The data can be used for educational and research purposes, and the original source will be properly acknowledged throughout the project.

Although privacy concerns are minimal, ethical considerations still apply. The analysis will be limited to the information available in the dataset, and any findings will be interpreted within the context of the available data. Since the dataset does not include factors such as driving behaviour, traffic conditions, weather, or vehicle maintenance history, conclusions about vehicle emissions will be based only on the variables included in the dataset.

To support reliable results, the dataset will be reviewed for missing values, duplicate records, and other data quality issues before modelling. Any preprocessing decisions will be documented, and model outputs will be interpreted carefully to avoid drawing conclusions that are not supported by the data.

4. Proposed Technical Pipeline

First, the dataset will be loaded into Python using Google Colab and examined for missing values, duplicate records, inconsistent entries, and other data quality issues. Missing values identified in the CO_2 Rating and Smog Rating variables will be addressed through an appropriate imputation strategy during preprocessing. The dataset will then be prepared for exploratory analysis and modeling.

The correlations between vehicle attributes, fuel consumption metrics, and CO2 emissions will then be better understood through an Exploratory Data Analysis (EDA) employing summary statistics, histograms, boxplots, scatterplots, and correlation analysis. This phase will help identify trends, patterns, and potential predictors relevant to the study questions.

After EDA, feature engineering and data preparation will be performed. Various categorical variables such as vehicle class, transmission type, fuel type, and manufacturer will be encoded into model-ready features. Numerical variables will be reviewed and prepared for modeling.

The prepared dataset will then be divided into training and testing datasets. In this project, Multiple Linear Regression will be used as the baseline model because of its simplicity as well as interpretability. Random Forest Regression will be used as a comparison model to evaluate whether a non-linear machine learning approach can improve predictive performance.

Model performance will be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics will be used to compare prediction accuracy, prediction error, and the models' ability to explain variation in CO_2 emissions.

Finally, the results will be reviewed to identify which vehicle features are most closely linked to fuel consumption and emissions, and to provide insights into transportation sustainability.

4.1 How the Stages Work Together

The stages of this project are designed to work together as a structured analytical process. Data profiling and preprocessing are completed first to ensure that the dataset is reliable and suitable for analysis. This provides a solid foundation for the exploratory analysis stage.

The insights gained from EDA help identify important patterns and relationships within the data, which in turn support feature selection and preparation for modelling. Once the data has been transformed into an appropriate format, predictive models can be developed and evaluated.

The final stage brings together the results from both the exploratory analysis and modelling processes. This makes it possible to answer the research questions, identify the factors most closely associated with fuel consumption and CO_2 emissions, and evaluate how effectively these factors can be used for prediction.

4.2 Feasibility Within Course Timeline

This project can be completed within the course timeline because the dataset is easy to access and is already organized. The dataset is not too large, making it suitable for analysis and predictive modelling. Also, tools and techniques used in this project are related to topics learned in the program courses. The project can be completed using Python and common analytics libraries without using advanced computing resources.

5. GenAI Declaration:

Generative AI tools may be used during this project to assist with learning, brainstorming, coding support, grammar review, and project organization. AI may also be used to help understand analytical concepts and troubleshoot technical issues encountered during the project.

Any information, code, or explanations generated by AI will be checked against course materials, project outputs, and relevant documentation before being used in the project.

One limitation of AI tools is that they may occasionally provide incorrect code, unsuitable modelling recommendations, or inaccurate interpretations of analytical results. For this reason, all preprocessing decisions, model outputs, and findings will be reviewed carefully before being included in the project.

While AI may be used as a supporting tool, all decisions related to data preparation, analysis, modelling, interpretation, and reporting will remain my responsibility. I will be able to explain and justify all work submitted as part of this project.

GenAI Risks, Impacts, and Mitigation Steps:

Risk	Impacts	Mitigation Steps
Incorrect AI-generated code	Wrong preprocessing or modelling results	Test and verify all code outputs before use

Unsuitable model suggestions	Using methods that do not fit the dataset	Select models based on course concepts and dataset characteristics
Misleading interpretation of results	Incorrect conclusions about emissions patterns	Compare findings with dataset trends and analytical evidence

References

Natural Resources Canada. (2026). *Fuel consumption ratings (2015-2024)*. Government of Canada Open Data Portal. https://open.canada.ca/data/en/dataset/98f1a129-f628-4ce4-b24d-6f16bf24dd64

Milestone 2

6. The Methodological Rationale (Literature Review)

6.1 Literature Review

Vehicle fuel consumption and CO_2 emissions have been widely studied because transportation remains a major contributor to greenhouse gas emissions. Researchers have used a variety of statistical and machine-learning techniques to better understand the factors that influence vehicle emissions and to improve prediction accuracy.

A common starting point in vehicle emissions research is Multiple Linear Regression because it is simple to implement and provides interpretable results. Regression-based methods help determine specific attributes that affect fuel consumption and emissions. Taşkın and Marttin (2023) demonstrated that regression methods can provide effective baseline predictions, making them useful for comparison against more advanced techniques. However, vehicle emissions are often influenced by multiple interacting factors, and these relationships are not always linear. As a result, traditional regression models may not fully capture the complexity of vehicle emissions data.

To address these limitations, many researchers have adopted machine learning techniques that can model more complex relationships. Ibrahim and Zakzouk (2026) found that machine learning methods are better suited for identifying patterns within vehicle emissions datasets because they are not restricted by strict linear assumptions. Similarly, Zahid and Jamil (2025) reported that Random Forest achieved strong predictive performance for fuel economy prediction tasks. These results suggest that machine learning approaches may provide more accurate predictions when vehicle characteristics interact in complex ways.

The fact that no single model consistently performs best across all datasets is another significant finding from the literature. To find the best model for a given problem, researchers frequently compare several methods using evaluation metrics such as RMSE, MAE, and R2. While Hamed et al. (2021) showed that Support Vector Machine models may also achieve significant predictive performance for transportation-related prediction tasks, Turhan et al. (2026) emphasised the significance of striking a balance between prediction accuracy and model interpretability.

Recent studies have also emphasized the importance of understanding which factors contribute most to emissions outcomes. Using explainable machine learning techniques, Yuan et al. (2025) identified fuel consumption measures, engine size, and engine configuration as some of the most influential variables affecting vehicle emissions. These findings highlight the necessity of analyzing both predictive performance and variable importance when studying vehicle environmental performance.

Overall, the studies reviewed indicate that both traditional regression methods and machine learning approaches can be effective for predicting vehicle emissions goals. Choice of model mainly depends on the complexity of the data and the objectives of the analysis. These findings therefore support the decision to compare Multiple Linear Regression and Random Forest Regression in this project.

6.2 Gap Analysis and justification

After reviewing the literature, I noticed that Multiple Linear Regression is commonly used as a baseline approach for predicting vehicle fuel consumption and CO_2 emissions. This is mainly because the model is simple to implement and the results are relatively easy to interpret. However, several studies also suggest that vehicle emissions are influenced by multiple factors that may interact with one another.

The dataset selected for this capstone project includes a mix of various vehicle characteristics, which include engine size, cylinder count, fuel type, transmission type, and vehicle class. These variables are most likely related to each other, and their impact on emissions may not always follow a simple linear pattern. As a result, a traditional regression model may not capture all of the relationships present in the data.

Furthermore, rather than depending just on one modelling strategy, a large number of the examined studies contrasted several modelling approaches. This suggests that assessing many models can help determine which strategy is best for a given dataset.

The main objective of this project is to assess the vehicle features with their CO2 emissions by comparing Random Forest Regression to Multiple Linear Regression. As a result, Multiple Linear Regression is used as the baseline model, as it provides a useful baseline benchmark. And Random Forest Regression was selected because it can handle more complex relationships and variable interactions without requiring strict linear assumptions.

Overall, this approach allows me to evaluate whether a machine-learning model can improve prediction performance while still focusing on maintaining a meaningful comparison with a commonly used baseline method. To ascertain whether the method is more appropriate for forecasting car emissions in the chosen Canadian dataset, model performance will be evaluated using evaluation metrics like MAE, RMSE, and R2.

7. Data Profiling and Audit

7.1 Dataset Overview and Structure

The dataset used in this capstone project contains fuel consumption and CO_2 emissions information for vehicles sold in Canada between 2015 and 2024. The data was obtained from the Government of Canada Fuel Consumption Ratings database and is available in CSV format. Each row in the dataset represents a vehicle model, while each column contains information about vehicle specifications, fuel consumption, emissions, or environmental ratings.

The dataset is structured data because it is organized in a tabular format with rows and columns. Therefore, no additional processing was required to convert the data into a usable format. The dataset contains both numerical variables, such as engine size and fuel consumption, and categorical variables, such as fuel type and vehicle class.

Dataset Characteristic	**Value**
Total Records	10,060
Total Variables	15

Numerical Variables	10
Categorical Variables	5
Time Period	2015–2024

After completing these preparatory procedures, the dataset was deemed appropriate for exploratory analysis and predictive modelling since it included complete observations for every variable.

7.2 Data Quality Assessment and Preparation

To find possible problems that might compromise the accuracy of the analysis and subsequent modelling, a data quality evaluation was carried out. The evaluation concentrated on data type verification, duplicate records, and missing values.

Two variables, CO2 Rating and Smog Rating, were missing values during the missing-value analysis. There were 1,128 missing values (11.2% of records) in the CO2 Rating variable and 2,234 missing values (22.2% of entries) in the Smog Rating variable. Both variables were kept in the dataset since they are environmental rating measures that could yield valuable information during analysis. To retain all vehicle records, median imputation was applied to both variables. This approach helped address the missing values while minimizing the influence of unusually high or low ratings.

A duplicate record check was also performed to identify repeated observations that could distort summary statistics or model performance. No duplicate records were found in the dataset. In addition, all variables were reviewed to ensure that numerical variables were stored using appropriate numeric data types and categorical variables were correctly represented.

Preparation Step	**Action Taken**
Missing Value Assessment	Identified missing values in CO2 Rating and Smog Rating
Missing Value Treatment	Applied median imputation to CO2 Rating and Smog Rating
Duplicate Check	No duplicate records found
Data Type Verification	Confirmed numerical and categorical variables
Dataset Validation	Verified complete observations after preprocessing

After preprocessing, all variables contained complete observations and were retained for exploratory analysis and modelling.

7.3 Univariate and Bivariate Profiling

To better understand the dataset, I carried out both univariate and bivariate analysis. The purpose of this step was to explore the distribution of individual variables and examine how different vehicle characteristics are related to fuel consumption and CO_2 emissions.

Univariate analysis

Descriptive statistics, boxplots, histograms, and frequency distributions were used in univariate analysis to comprehend the properties of individual variables. Significant differences in vehicle specifications and emissions levels were found throughout the sample.

Key Univariate Findings

Analysis Area	Observation
Engine Size	Ranged from 0.9 L to 8.4 L with an average of 3.15 L
Cylinders	Varied from 3 to 16 cylinders with an average of 5.62
Combined Fuel Consumption	Ranged from 4.0 to 26.1 L/100 km with an average of 10.98 L/100 km
CO_2 Emissions	Ranged from 94 to 608 g/km with an average of 253.71 g/km
Outliers	Several high-emission and high- consumption vehicles were identified.

The fuel consumption boxplot revealed several outliers among vehicles with extremely high fuel consumption levels. These observations were retained because they represent valid vehicle categories rather than data entry errors.

Bivariate Analysis

Following the univariate analysis, bivariate analysis was conducted to examine relationships among variables using boxplots, scatterplots, and correlation analysis.

According to the visualizations, cars with bigger engines often used more gasoline and emitted more CO2. Cylinder count showed similar patterns: cars with more cylinders typically used more fuel. Distinctions between fuel kinds and vehicle classes were also apparent.

Key Bivariate Findings

Variable Relationship	Observation
Engine Size vs CO_2 Emissions	Strong positive relationship
Cylinders vs Fuel Consumption	Fuel consumption increased with cylinder count

Vehicle Class vs CO_2 Emissions	Larger vehicle classes produced higher emissions
Fuel Type vs CO_2 Emissions	Emissions varied across fuel types
Fuel Consumption vs CO_2 Emissions	Strong positive relationship

While analyzing the correlation matrix, the following findings were noted. While performing EDA, Combined fuel consumption represents the strongest relationship with CO_2 emissions ($r = 0.97$), followed by city fuel consumption ($r = 0.94$) and highway fuel consumption ($r = 0.91$). Engine size ($r = 0.84$) and cylinder count ($r = 0.82$) were also strongly related to emissions.

These results suggest that fuel consumption and engine-related variables are likely to be important predictors in the modelling stage of the project.

Comparative Analysis Findings

Finding	Result
Highest Average CO2 Emission Vehicle Class	Van: Passenger (389.7 g/km)
Second Highest CO2 Emission Vehicle Class	Sport Utility Vehicle: Standard (303.9 g/km)
Highest Fuel Consumption Fuel Type	Ethanol (16.63 L/100 km)
Lowest Fuel Consumption Fuel Type	Diesel (9.34 L/100 km)

The results indicate that fuel consumption and engine-related variables are likely to be the most influential predictors of vehicle emissions and should play a central role in subsequent modeling stages.

7.4 Dataset Constraints and Modelling Implications

Constraint Identified	Impact on Analysis and Claims
Missing values in CO2 Rating and Smog Rating	Missing values were addressed using median imputation. While this allowed the variables to be retained, imputed values may not fully reflect the original ratings.
Dataset represents vehicles available in the Canadian market	Results may not generalize to vehicle fleets, regulations, or driving conditions in other countries.
Presence of outliers	Extreme vehicle categories may influence summary statistics and model performance.

Strong correlations among fuel variables	Several fuel-consumption variables contain overlapping information, making it difficult to isolate the individual contribution of each predictor.
Dataset contains vehicle specifications only	Conclusions can be made about relationships between vehicle characteristics and emissions, but not about driver behaviour, maintenance practices, or real-world operating conditions.

Overall, the dataset provides sufficient information to analyze vehicle fuel consumption and CO_2 emissions. However, the dataset contains vehicle specifications and fuel consumption measurements but does not include factors such as driving behavior, traffic conditions, weather, or vehicle maintenance, all of which may influence real-world emissions.

These limitations may affect the modelling stage because the models can only learn from the variables available in the dataset. As a result, the predictions will reflect relationships between vehicle characteristics and emissions rather than all factors that influence vehicle performance. Therefore, the modelling results should be interpreted within the scope of the available data and not as a complete representation of real-world driving conditions.

7.5 Privacy and Ethical Considerations

The dataset used in this project contains only vehicle specifications, fuel consumption measurements, and emissions information. It does not include any personally identifiable information such as names, addresses, phone numbers, or other personal records.

Since the dataset does not involve individuals or human participants, privacy concerns are minimal. The project also does not involve surveys, interviews, or human-subject research. For this reason, the Tri-Council Policy Statement (TCPS 2) is not directly applicable to this study.

Although privacy risks are low, the analysis will still follow responsible data management practices. Findings will be reported accurately and interpreted within the limitations of the dataset to avoid unsupported conclusions.

8. Architecture and Logic: The Project Approach

8.1 System Architecture and Workflow

Figure 8.1 shows the overall workflow that will be followed throughout this project. The process begins with the Government of Canada Fuel Consumption Ratings dataset, which contains information about vehicle specifications, fuel consumption, environmental ratings, and CO_2 emissions for vehicles sold in Canada between 2015 and 2024.

Figure 8.1: System Architecture and Analytical Workflow

Examining and comprehending the dataset through data profiling and exploratory data analysis (EDA) is the initial stage of the procedure. In order to better understand the connections between car attributes, fuel consumption, and emissions, I look at the dataset structure, find missing values, produce summary statistics, and make visualisations. Before modelling starts, this preliminary evaluation gives me a better grasp of the variables and helps me spot possible problems with data quality.

After the initial analysis, preprocessing is performed to prepare the dataset for modeling. Missing values are addressed and imputed, data types are verified, duplicate records are identified, and essential outliers are reviewed. These steps help me to improve the data quality and ensure consistency throughout the analysis.

The next stage focuses on feature engineering. The target variable, CO_2 Emissions (g/km), is identified, and relevant predictor variables are selected. Categorical variables such as vehicle class, transmission type, fuel type, make, and model are converted into numerical features using one-hot encoding so they can be used in machine-learning models.

The dataset is divided into training and test sets at 80/20 when it is ready for modelling. The testing dataset is set aside to assess model performance on untested data, whereas the training dataset is used to build the predictive models

Two regression models will be developed and compared. Multiple Linear Regression will be used as the baseline model because it provides a simple and interpretable benchmark. Random Forest Regression will be used as the comparison model to determine whether a machine-learning approach can improve predictive performance. In addition, 5-fold cross-validation will be applied during model development to provide a more reliable assessment of model performance.

The final stage focuses on evaluating and interpreting the results. Model performance will be assessed using MAE, RMSE, and R^2 score. The findings will then be used to identify the vehicle characteristics that contribute most strongly to fuel consumption and CO_2 emissions and to determine which modelling approach performs best for the selected dataset.

8.2 Preprocessing (The Foundation)

Before developing the predictive models, the dataset was prepared to ensure that it was suitable for analysis. This stage focused on addressing data quality issues identified during the audit and creating a consistent dataset for subsequent modelling activities.

As discussed in Section 7.2, missing values were identified in the CO_2 Rating and Smog Rating variables and were addressed using median imputation. The dataset was also reviewed for duplicate records and data type consistency. No duplicate observations were found, and all variables were verified to ensure they were stored in the appropriate format for analysis.

In the exploratory analysis stage, extreme outliers were identified. Unusual relationships, such as high fuel consumption and emission levels, were observed in a number of vehicle categories; these observations were retained because they indicate legitimate vehicle categories rather than data entry errors. And retaining these records ensures the dataset reflects the full range of light-duty vehicles available for sale in the Canadian market.

By completing these preprocessing steps, the dataset was transformed into a clean and consistent form suitable for feature engineering and predictive modelling.

After preprocessing, the dataset contained complete observations and was considered suitable for feature engineering and model development.

8.3 Feature Engineering (The Transformation)

After preprocessing was completed, the next step involved preparing the variables for predictive modelling. Since the objective of this project is to predict vehicle CO_2 emissions, CO_2 Emissions (g/km) was selected as the target variable.

The predictor variables were chosen based on the findings from the exploratory analysis and the literature review. Variables such as engine size, cylinder count, vehicle class, transmission type, fuel type, fuel consumption measures, and Smog Rating were included because they are expected to have a direct relationship with vehicle emissions.

Machine-learning algorithms cannot directly use some of the chosen variables because they are categorical. In order to solve this, one-hot encoding will be used to transform categorical characteristics like vehicle class, gearbox type, fuel type, make, and model into numerical variables. The models are able to interpret categorical data while maintaining the distinctions between categories thanks to this transformation.

The final modelling dataset will be created by combining the encoded categorical variables with the numerical variables in their original format. A fair and consistent comparison between the models is ensured by using the same set of features for both Random Forest Regression and Multiple Linear Regression.

Once these transformations are completed, the dataset will be ready for model training and evaluation.

Feature Engineering Summary

Variable Type	Examples	Transformation Applied
Numerical Variables	Engine Size, Cylinders, Fuel Consumption Measures	Retained in numerical format
Categorical Variables	Make, Model, Vehicle Class, Transmission, Fuel Type	One-Hot Encoding
Target Variable	CO_2 Emissions (g/km)	Prediction Output
Predictor Variables	Vehicle Specifications and Fuel Consumption Variables, and Smog Rating	Used as Model Inputs

After feature engineering, the dataset will contain numerical predictor variables that can be used directly by machine-learning algorithms. The same engineered features will be used for both models to ensure a fair comparison.

8.4 Modelling and Evaluation Strategy

Model Inputs and Outputs

The purpose of the modeling stage is to predict vehicle CO_2 emissions using vehicle specifications and fuel-consumption characteristics.

The model output will be a predicted value for CO_2 Emissions (g/km). The resulting predictions will help answer the project research questions by identifying which vehicle characteristics are associated with higher emissions and by evaluating how accurately emissions can be predicted using vehicle specifications.

Model Development Approach

To answer the research questions, two regression models will be developed and compared. Multiple Linear Regression will be used as the baseline model because it is widely used, easy to interpret, and provides a useful benchmark for comparison. Random Forest Regression will also be developed to determine whether a machine-learning approach can improve prediction accuracy by capturing more complex relationships within the data.

Comparing these two models will help determine whether a more complex machine-learning approach provides meaningful improvements in prediction accuracy.

Training and Validation Strategy

The dataset will be divided into training and testing datasets using an 80/20 split. The training dataset will be used to develop the models, while the testing dataset will be reserved for evaluating performance on unseen observations.

During model training, 5-fold cross-validation will be used to improve the reliability of the results. Instead of relying on a single train-test split, this method enables models to be assessed across several subsets of the training data. Consequently, it is anticipated that the performance estimates will be less reliant on a specific data division and more consistent.

The testing dataset will be used during the final evaluation and will not be used during model training.

Leakage Prevention and Fair Comparison

This prevents information from the testing dataset from influencing model development and ensures that performance estimates remain unbiased.

Both Multiple Linear Regression and Random Forest Regression will use the same training dataset, testing dataset, predictor variables, and evaluation metrics. This ensures that any performance differences observed between models are caused by the modeling techniques themselves rather than differences in data preparation.

Evaluation Metrics

Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score will be used to assess the model's performance because the goal is to predict a continuous numerical value.

MAE measures the average prediction error, RMSE places greater emphasis on larger prediction errors, and R^2 measures the proportion of variation in CO_2 emissions explained by the model. These metrics provide complementary measures of prediction accuracy and model fit and are commonly used for regression-based prediction problems (Natarajan et al., 2023).

The best model for this project will be the one that performs the best in terms of prediction while still being reasonably interpretable.

9. Reproducibility and Version Control

To support reproducibility and version control, this project is maintained in a GitHub repository that contains all project files, documentation, notebooks, and generated outputs. The repository is organized using a clear folder structure to allow a third-party reviewer to easily navigate the project and reproduce the analysis workflow.

The repository includes:

- A README.md file that describes the project problem, dataset source, methodology, and execution steps.
- A Dataset folder containing the original vehicle emissions dataset.
- A Notebooks folder containing the Google Colab notebooks used for profiling, preprocessing, feature engineering, modelling, and evaluation.

- EDA Reports folder containing automated profiling reports and manual exploratory data analysis outputs used during data profiling and audit.

The GitHub repository can be accessed at:

GitHub Repository: https://github.com/IndrajeetSingh121/CIND820-CO2-Emission-Prediction

10. AI Use Declaration

During this stage of the project, generative AI tools were used to learn how to use GitHub for repository creation, file organization, and README development, as I had no previous experience with GitHub. AI assistance was also used to learn how to generate and export HTML profiling reports from Google Colab notebooks.

The information and instructions provided by AI were verified through testing, successful implementation, and comparison with official documentation before being applied to the project. All project decisions, data analysis, preprocessing methods, modelling choices, interpretations, and written content were reviewed and verified by me before inclusion in the report.

I remain responsible for all submitted work and can explain and justify the methods, results, and conclusions presented in this project.

References

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